AEL: Evolving Agent Harness in Open-Ended Environments

Wujiang Xu, Jiaojiao Han, Minghao Guo, Kai Mei, Xi Zhu, Han Zhang, Dimitris N. Metaxas · Rutgers University · arXiv 2604.21725v2

HarnessRSI 74 ★★★

evidence: improver

Paper

Code

1 What changes, what is kept

Frozen LLM; learned part of the harness = memory use

- **Learned** retrieval-policy bandit (Thompson)
- **Learned** reflection insight in planner prompt
- **Learned** procedural rules from 3-tier memory
- **Fixed** planner, 12 tools, LLM (default)

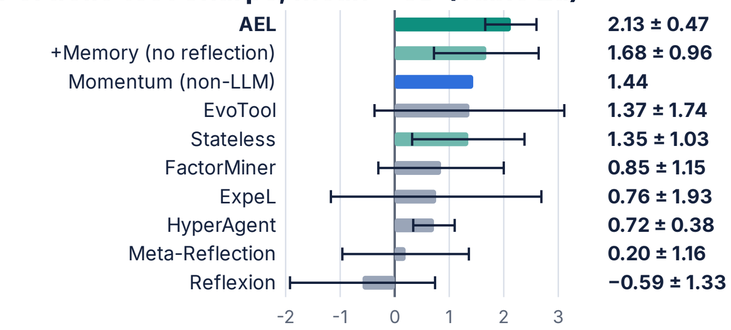
1. **Policy pool** — 5 starting retrieval policies (from none to aggressive recall), each with a Beta posterior
2. **Thompson sample, retrieve** — play the top sample; recall from episodic / semantic / procedural memory
3. **Act, score, update** — frozen LLM + fixed planner and tools act; reward r = (s+1)/2 updates the posterior
4. **Slow: LLM reflection** — diagnoses why the window went well or badly; insight goes into the planner prompt
5. **Add an arm if the pool is weak** — mean reward < 0.4: LLM writes a new retrieval policy (AST-checked in App P.2); Beta(1, 1)

chronological split: posteriors picked on validation
test: all learning off (frozen bandit, read-only memory)

green loop: fast, every episode (steps 1–3)
purple loop: slow, every slow window; arm added only if pool mean reward < 0.4 (steps 4–5)

2 Headline: best mean on both benchmarks

Portfolio test Sharpe, mean ± SD (Table 2a)



Claude Haiku 4.5, 5 seeds, 28 test bars. Whiskers ±1 SD. Momentum = deterministic non-LLM. AEL also leads Sortino (4.08) and Calmar (10.40). Routing results (Gemini 3.1 Flash-Lite) are in the table under 4.

3 Controls that decide if the gain is real

Ablation · test Sharpe (Table 3, selected rows, no SDs)



Welch p, AEL vs	
p < .05	Reflexion .008 · Meta-R .017 · HyperAgent .001
p > .05	EvoTool .393 · +Memory .384 · +Tools .200 · ExpeL .190 · Stateless .178 · FactorMiner .066

Mechanism given to baselines? No: authors' adaptations without a policy bandit or arm injection; no matched-budget row. **Cost:** ~$2.80 per run (App E); "evolution cost" ~400 LLM calls vs ~20 for HyperAgent (Tab 16); baseline costs not given. Headline at 0 bp cost; ~20 bp: AEL ~1.3, momentum ~1.0, EvoTool ~0.7 (App R).

4 Where the gain is absent or thin

- **Not significant** against the strongest comparators (EvoTool p = .393, +Memory .384)
- Max drawdown: EvoTool −1.46 and equal-weight −1.47 beat AEL −1.53; momentum baseline (1.44): no test vs AEL
- Routing: fixed class-balanced 0.565 beats every prior baseline; finance test = 28 bars; coding / web not run
- Paper inconsistencies: HyperAgent 0.72 (Tab 2a) vs 0.46 (App K); LLM-credit 1.37 / 1.49 / −0.53; "4 of 9" significant, 3 named

Other streams (5 seeds)	AEL	no reflection	best other
Routing, LLM agent (Tab 2b)	0.840	0.711	0.565 fixed
last regime only	0.885	0.685	0.705 prior
Synthetic bandit, last regime	0.452	0.359	0.410 UCB1
20Newsgroups (App A.4)	0.253	0.254	0.271 fixed

LLM run: no-reflection already 0.685 in the last regime (0.206 in the deterministic stream, Tab 11). Synthetic arm injection is scripted. 20Newsgroups: no lift (p = 0.97).

Takeaway

A Thompson-sampling bandit over retrieval policies, plus LLM reflection that can add one, has the best mean on both benchmarks (Sharpe 2.13; routing accuracy 0.840). The mean gain over the strongest comparators is not significant at 5 seeds; the lower variance is significant against 2 of 9 baselines (ExpeL, EvoTool) and the shuffled-reflection drop (1.54) has no test. No lift where the best policy is stable.

Our read · HarnessRSI

Score 74 · ★★★ · evidence level: improver

Reports: chronological held-out test, 5 seeds (SD), ablations incl. shuffled reflection, AEL run cost.

Does not report: budget-matched baselines, transfer of an evolved artifact, leakage test, baseline costs.